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Subject : Applied Physics
Assignment no : 1

Sample Problem 25.2

$q_1 = q_2 = 1.37 \times 10^{5} C$

$r = 100 m$

$F = ?$

$$F = \frac{k q_1 q_2}{r^2}$$

$$F = \frac{9\times10^{9} \times 1.37\times10^{5} \times 1.37\times10^{5}}{(100)^2}$$

$$\boxed{F = 1.69 \times 10^{16}\ N}$$

Sample problem 25.3

$q_1 = q_2 = 1.6 \times 10^{-19} C$

$r = 5.3 \times 10^{-11} m$

a) $F = ?$ electrostatic

$$F_e = \frac{k q_1 q_2}{r^2}$$

$$F_e = \frac{9\times10^{9} \times (1.6\times10^{-19})^2}{(5.3\times10^{-11})^2}$$

$$\boxed{F_e = 8.2 \times 10^{-8}\ N}$$

b) $F_g = \frac{G m_1 m_2}{r^2}$

$F_g = \frac{6.67 \times 10^{-11} \times 9.111 \times 10^{-31} \times 1.67 \times 10^{-27}}{5.3 \times 10^{-11}}$

$\boxed{F_g = 3.6 \times 10^{-47} N}$

Sample Problem 25.4

$q_1 = q_2 = 1.6 \times 10^{-19}$

$r = 4 \times 10^{-15} m$

$F = ?$

$F = \frac{1}{4\pi\varepsilon_0} \frac{e^2}{r^2}$

$F_e = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(4 \times 10^{-15})^2}$

$\boxed{F_e = 14 N}$

Sample Problem 26.1

$F_y = 3.6 \times 10^{-8} N$

$q_e = -1.6 \times 10^{-19}$

a) $E_y = ?$

b) $q_\alpha = ?$

$F = ?$

a) $E_y = \frac{F_y}{q} = \frac{3.6 \times 10^{-8}}{-1.6 \times 10^{-19}} = -2.25 \times 10^{11} \, NC^{-1}$

b) $F_y = 2qE \qquad q_\alpha = 2q_e$

$= 2(1.6 \times 10^{-19})(-2.25 \times 10^{11})$

$= -72 \times 10^{-8} \, N$

The force is in -ve direction.

Sample Problem 26.2

$q_e = 1.6 \times 10^{-19}$

$r = 26.5 \times 10^{-12} m$

$E = ?$

$E = \frac{1}{4\pi\varepsilon_0} \frac{|q|}{r^2}$

$= \frac{9 \times 10^9 (2 \times 1.6 \times 10^{-19})}{26.5 \times 10^{-12}} \qquad \therefore q_{He} = 2e$

$= 4.1 \times 10^{12} \, NC^{-1}$

Sample Problem 26.3

$q_1 = +1.5 \times 10^{-6} C$

$q_2 = +2.3 \times 10^{-6} C$

$L = 13 cm$

Point where $E = 0$?

$\frac{\cancel{1} \, q_1}{\cancel{4\pi}\varepsilon_0 x^2} = \frac{\cancel{1} \, q_2}{\cancel{4\pi}\varepsilon_0 (13 - x)^2}$

$169 - 26x - x^2 = \frac{q^2 \, x^2}{q_1}$

$$q_1(13-x)^2 = q_2 x^2$$

taking sqrt on both sides.

$$\sqrt{q_1(13-x)^2} = \sqrt{q_2 x^2}$$

$$\sqrt{q_1}\,(13-x)^2 = \sqrt{q_2}\,x$$

$$13-x = \sqrt{\frac{q_2}{q_1}}\,x$$

$$x = \frac{13}{1 \pm \sqrt{q_2/q_1}}$$

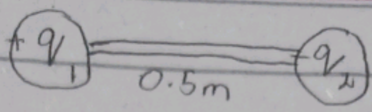
$$x = 5.8\,\text{cm}$$

or

$$x = -54\,\text{cm}$$

25.1

$r = 50\,cm$



$r = 0.5\,m$

$F_1 = 0.108\,N$

$$F_1 = \frac{k q_1 q_2}{r^2}$$

$$0.108 = \frac{9\times10^9 q_1 q_2}{(0.5)^2}$$

$$q_1 q_2 = \frac{0.108 \times (0.5)^2}{9\times10^9}$$

$$q_1 q_2 = 3\times10^{-12} \quad \text{— (i}$$

$$F_2 = 0.036\,N$$

$$q = \frac{q_1 - q_2}{2}$$

$$F_2 = \frac{(K)\left(\frac{q_1 - q_2}{2}\right)\left(\frac{q_1 - q_2}{2}\right)}{r^2}$$

$$F_2 = \frac{K \frac{(q_1 - q_2)^2}{4}}{r^2}$$

$$F_2 \times r^2 \times 4 = K(q_1 - q_2)^2$$

$$\frac{F_2 \times r^2 \times 4}{K} = (q_1 - q_2)^2$$

$$\frac{0.036 \times (0.5)^2 \times 4}{9\times10^9} = (q_1 - q_2)^2$$

$$\boxed{q_1 - q_2 = 2 \times 10^{-6}}$$

$$(a+b)^2 = (a-b)^2 + 4ab$$

$$(q_1 + q_2)^2 = (q_1 - q_2)^2 + 4q_1q_2$$

$$(q_1 + q_2)^2 = (2 \times 10^{-6})^2 + 4 \times 3 \times 10^{-12}$$

$$(q_1 + q_2)^2 = 16 \times 10^{-12}$$

$$\boxed{q_1 + q_2 = 4 \times 10^{-6}} \text{ —(a)}$$

$$\boxed{q_1 - q_2 = 2 \times 10^{-6}} \text{ —(b)}$$

Add (a) and (b)

$$2q_1 = 6 \times 10^{-6}$$

$$\boxed{q_1 = 3 \times 10^{-6}}$$

$$3 \times 10^{-6} + q_2 = 4 \times 10^{-6}$$

$$\boxed{q_2 = 1 \times 10^{-6} \, C}$$